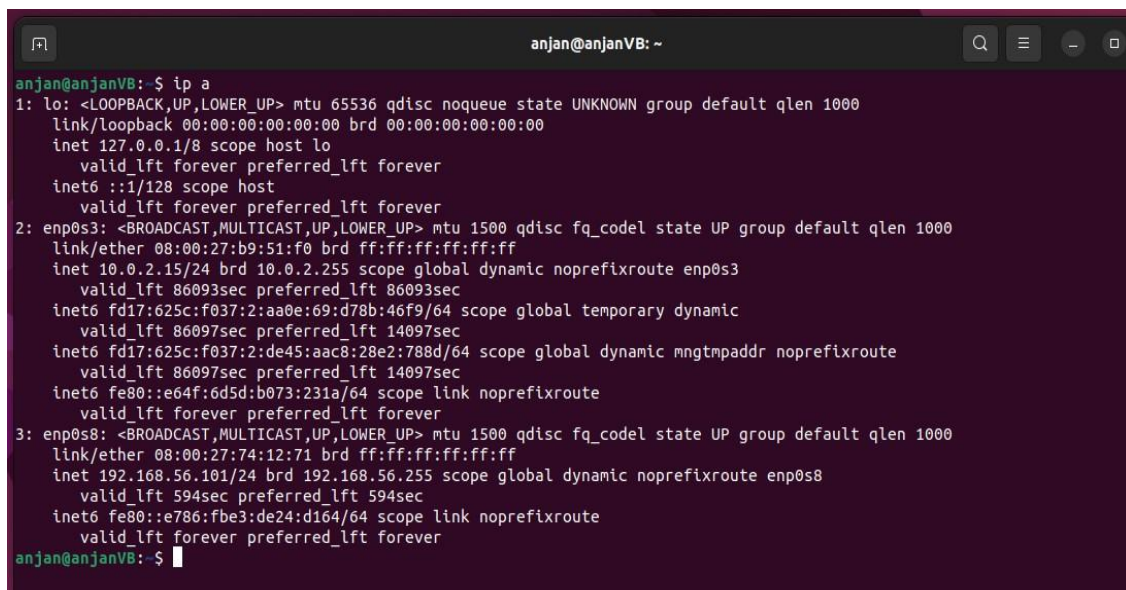


Implementation

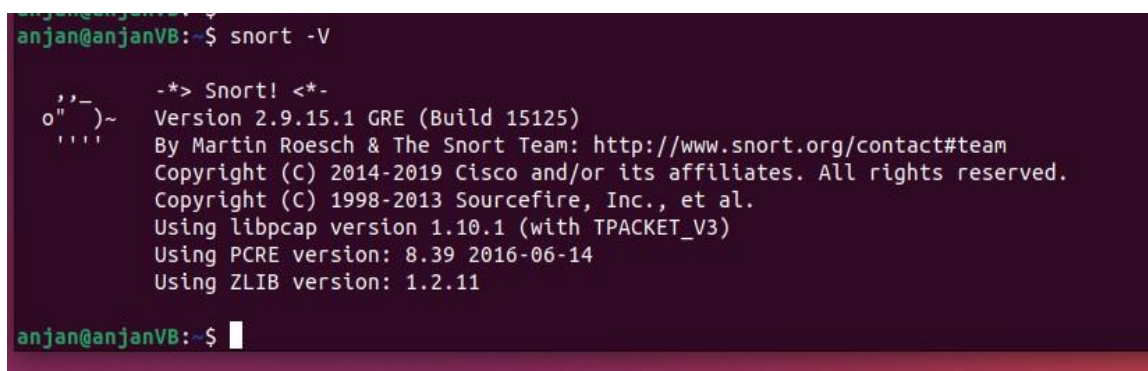
Environment Setup

The project was implemented using two virtual machines in Oracle VirtualBox. Ubuntu was used as the main IDS server where Snort, tcpdump, Python scripts, and the Streamlit dashboard were executed. Kali Linux was used as the attacker machine to generate network attacks and suspicious traffic. Both machines were connected using a Host-Only Adapter to create an isolated testing environment where attacks could be safely performed without affecting external networks. Required Python libraries such as pandas, scikit-learn, streamlit, and joblib were installed to support machine learning classification and dashboard development.



```
anjan@anjanVB: ~  
anjan@anjanVB:~$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:b9:51:f0 brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3  
        valid_lft 86093sec preferred_lft 86093sec  
    inet6 fd17:625c:f037:2:aa0e:69:d78b:46f9/64 scope global temporary dynamic  
        valid_lft 86097sec preferred_lft 14097sec  
    inet6 fd17:625c:f037:2:de45:aac8:28e2:788d/64 scope global dynamic mngtmpaddr noprefixroute  
        valid_lft 86097sec preferred_lft 14097sec  
    inet6 fe80::e64f:6d5d:b073:231a/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:74:12:71 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s8  
        valid_lft 594sec preferred_lft 594sec  
    inet6 fe80::e786:fbe3:de24:d164/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
anjan@anjanVB:~$
```

Figure 1: ubuntu setup



```
anjan@anjanVB:~$ snort -V  
    ,,-  
    o" )~  
    '  '  
    -*> Snort! <*-  
    Version 2.9.15.1 GRE (Build 15125)  
    By Martin Roesch & The Snort Team: http://www.snort.org/contact#team  
    Copyright (C) 2014-2019 Cisco and/or its affiliates. All rights reserved.  
    Copyright (C) 1998-2013 Sourcefire, Inc., et al.  
    Using libpcap version 1.10.1 (with TPACKET_V3)  
    Using PCRE version: 8.39 2016-06-14  
    Using ZLIB version: 1.2.11  
anjan@anjanVB:~$
```

Figure 2: snort installation

Traffic generation from kali linux

To test the IDS system, attack traffic was generated from Kali Linux using tools such as Nmap and Ping. These attacks simulate suspicious and malicious activities such as reconnaissance scans and network flooding. This generated traffic acts as the input for Snort detection and machine learning classification.

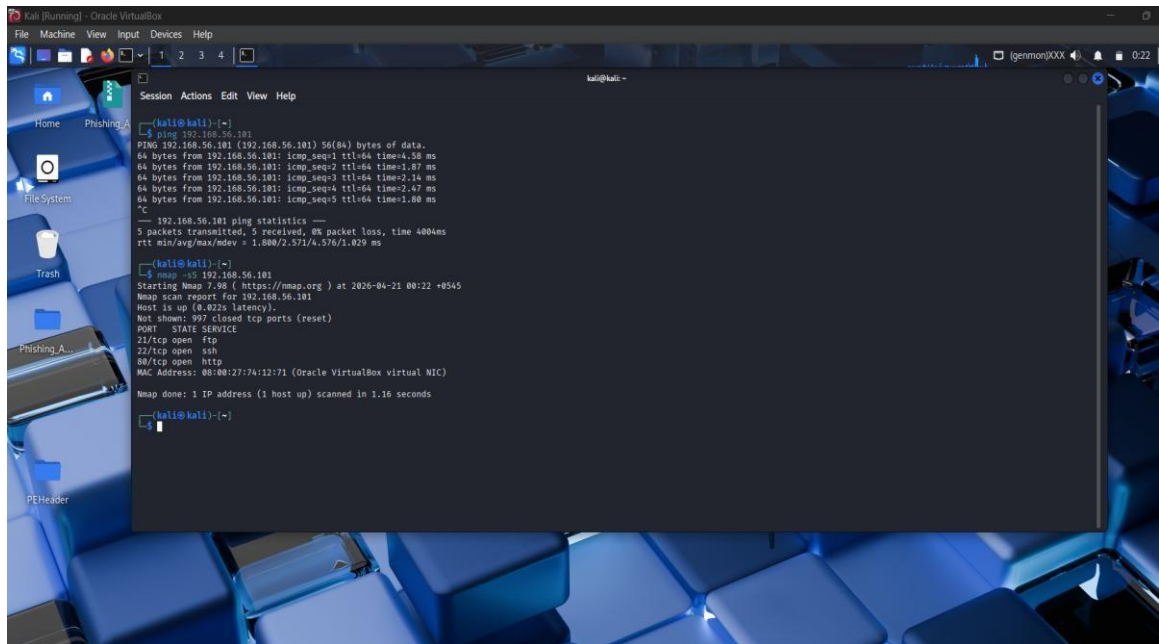


Figure 3: generating traffic from kali linux

Snort IDS Detection

Snort was configured as the primary signature-based intrusion detection system. It continuously monitors incoming traffic and compares packets against predefined rule sets. When known attack patterns are detected, Snort generates alerts and stores them in the alert log file. This forms the first detection layer of the Hybrid IDS.

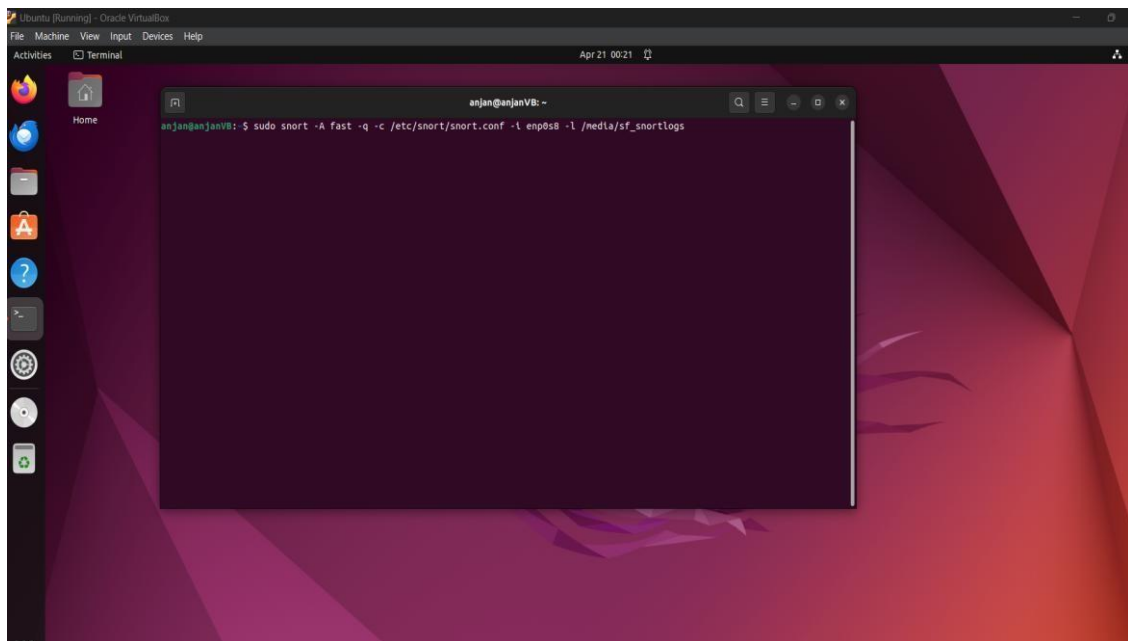


Figure 4: starting snort

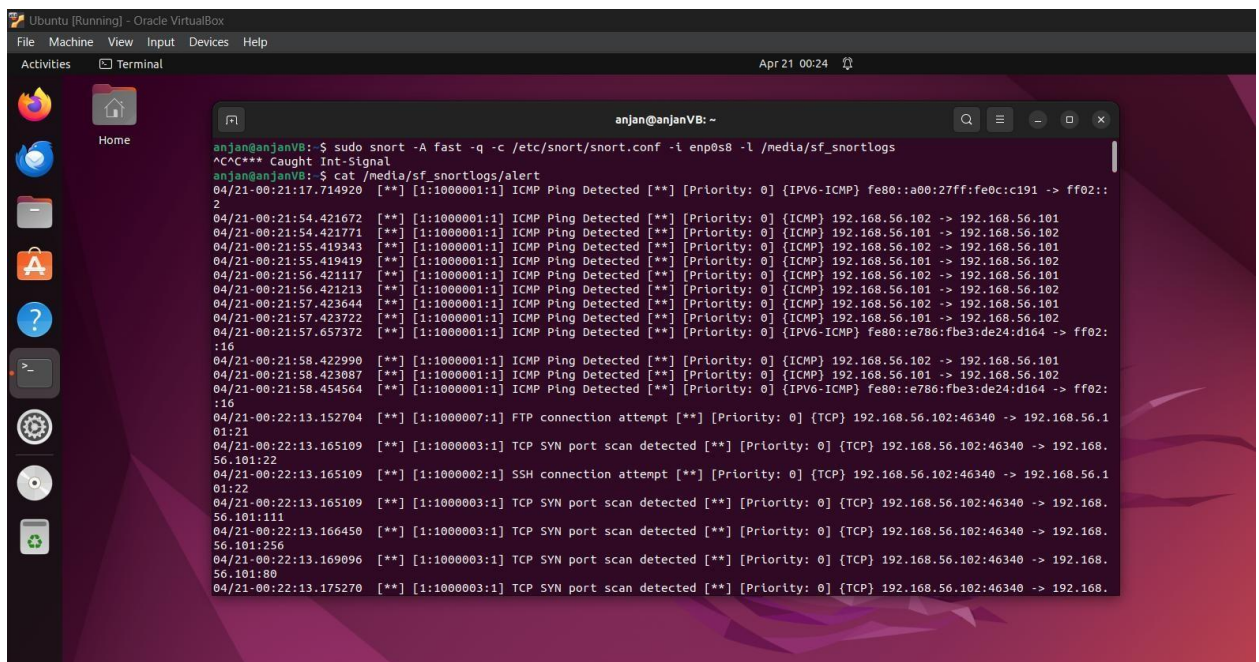


Figure 5: snort alert log

Machine Learning Model Development

A Random Forest machine learning model was used to improve detection beyond traditional rule-based IDS. The model processes Snort alerts and classifies them into four categories:

- Attacks
- Probe/Scan
- Normal
- Suspicious

This helps prioritize alerts and identify uncertain traffic that may require manual review.

```
PS C:\Users\neoteric\OneDrive\Desktop\Project> python hybrid_pipeline.py
=====
Hybrid IDS Pipeline – Snort + ML Analysis
Attack threshold: 50% Probe/Normal threshold: 80%
=====

[1/5] Loading model ...
Classes : ['high', 'low', 'medium']
Features: 51

[2/5] Reading Snort alerts:
C:\Users\neoteric\OneDrive\Desktop\Project\snortlogs\alert
Raw alert lines: 0

No alerts found. Check C:\Users\neoteric\OneDrive\Desktop\Project\snortlogs\alert
PS C:\Users\neoteric\OneDrive\Desktop\Project> python hybrid_pipeline.py
=====
Hybrid IDS Pipeline – Snort + ML Analysis
Attack threshold: 50% Probe/Normal threshold: 80%
=====

[1/5] Loading model ...
Classes : ['high', 'low', 'medium']
Features: 51

[2/5] Reading Snort alerts:
C:\Users\neoteric\OneDrive\Desktop\Project\snortlogs\alert
Raw alert lines: 2,663

[3/5] Aggregating repeated alerts into unique events ...
2,663 raw lines → 12 unique events

[4/5] Running ML predictions ...
[001] 04/21-00:50:21.822376 | ● SUSPICIOUS (67%) | x1 | ICMP Ping Detected
[002] 04/21-00:50:42.808045 | ● ATTACK (67%) | x1 | Telnet connection attempt
[003] 04/21-00:50:42.815573 | ● ATTACK (66%) | x18 | FTP connection attempt
[004] 04/21-00:50:42.820537 | ● SUSPICIOUS (61%) | x10 | SSH connection attempt
[005] 04/21-00:50:42.826241 | ● PROBE/SCAN (100%) | x1,041 | TCP SYN port scan detected
[006] 04/21-00:50:43.104696 | ● SUSPICIOUS (61%) | x1 | SNMP AgentX/tcp request
```

Figure 6: hybrid pipeline(1)

```

[3/5] Aggregating repeated alerts into unique events ...
      2,663 raw lines → 12 unique events

[4/5] Running ML predictions ...
[001] 04/21-00:50:21.822376 | ● SUSPICIOUS (67%) | x1 | ICMP Ping Detected
[002] 04/21-00:50:42.808045 | ● ATTACK (67%) | x1 | Telnet connection attempt
[003] 04/21-00:50:42.815573 | ● ATTACK (66%) | x18 | FTP connection attempt
[004] 04/21-00:50:42.820537 | ● SUSPICIOUS (61%) | x10 | SSH connection attempt
[005] 04/21-00:50:42.826241 | ● PROBE/SCAN (100%) | x1,041 | TCP SYN port scan detected
[006] 04/21-00:50:43.104696 | ● SUSPICIOUS (61%) | x1 | SNMP AgentX/tcp request
[007] 04/21-00:50:43.265484 | ● SUSPICIOUS (61%) | x1 | SNMP request tcp
[008] 04/21-00:50:50.514856 | ● ATTACK (67%) | x788 | ICMP Ping Detected
[009] 04/21-00:50:50.514976 | ● ATTACK (67%) | x790 | ICMP Ping Detected
[010] 04/21-00:50:50.617067 | ● PROBE/SCAN (100%) | x4 | Nmap NULL scan detected
[011] 04/21-00:50:50.745320 | ● PROBE/SCAN (100%) | x4 | Nmap XMAS scan detected
[012] 04/21-00:50:50.745320 | ● PROBE/SCAN (100%) | x4 | SCAN nmap XMAS

[5/5] Saving results:
      C:\Users\neoteric\OneDrive\Desktop\Project\hybrid_results.csv

=====
Raw alert lines   : 2,663
Unique events     : 12
=====
● ATTACK          : 4
● PROBE/SCAN      : 4
● NORMAL          : 0
● SUSPICIOUS      : 4
=====

=====
Pipeline complete!
=====

```

Figure 7: hybrid pipeline(2)

PCAP Analysis Implementation

Tcpdump was used to capture all network traffic into a PCAP file. The PCAP analyzer processes this file to reconstruct traffic flows and compare them against Snort alerts. This helps identify traffic that Snort failed to detect. These missed flows are then classified using the machine learning model to improve overall detection capability.

```
PS C:\Users\neoteric\OneDrive\Desktop\Project> python pcap_analyzer.py
=====
PCAP Analyzer – Snort Coverage vs ML Classification
Attack threshold: 50% Probe/Normal threshold: 80%
=====

[1/5] Model loaded. Classes: ['high', 'low', 'medium'] Features: 51

[2/5] Reading PCAP: C:\Users\neoteric\OneDrive\Desktop\Project\snortlogs\traffic.pcap
      Total packets: 4,252

[3/5] Building flows from 4,252 packets ...
      Unique flow signatures: 1,128

[4/5] Loading Snort signatures: C:\Users\neoteric\OneDrive\Desktop\Project\snortlogs\alert
      Snort alert signatures: 990

[5/5] Classifying 1,128 flows with ML ...
█
```

Ln 57, Col 45

Figure 8: pcap analyzer(1)

```
[5/5] Classifying 1,128 flows with ML ...

Raw packets parsed           : 4,252
Raw flow signatures          : 1,128
Unique events (after aggregation) : 49

✓ Snort detected events      : 38
⚠ Snort missed events       : 11

Of the Snort-missed events, ML says:
● ATTACK      : 0
● PROBE/SCAN  : 9
● NORMAL      : 0
● SUSPICIOUS  : 2

Saved: C:\Users\neoteric\OneDrive\Desktop\Project\pcap_analysis_results.csv

=====
PCAP analysis complete!
Ln 14
```

Figure 9: pcap analyzer(2)

Dashboard Implementation

A Streamlit dashboard was developed to provide a centralized interface for visualizing all IDS results. The dashboard displays:

- Snort alerts
- ML classification results
- Protocol distribution
- Top source IPs
- Snort coverage
- PCAP missed flow analysis

This improves usability by presenting technical detection results in a clear and understandable format.

```

PS C:\Users\neoteric\OneDrive\Desktop\Project> streamlit run dashboard.py

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://192.168.1.76:8501

2026-04-21 02:09:58.201 Please replace `use container width` with `width`.

```

Figure 10: dashboard running message

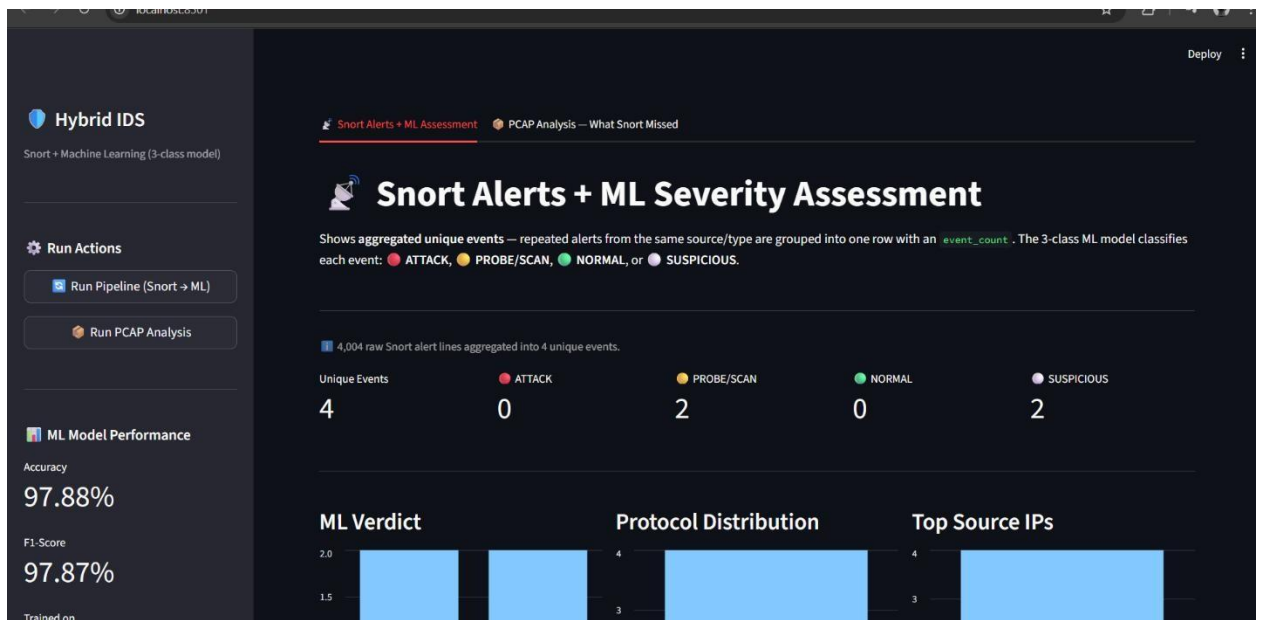


Figure 11: dashboard tab1

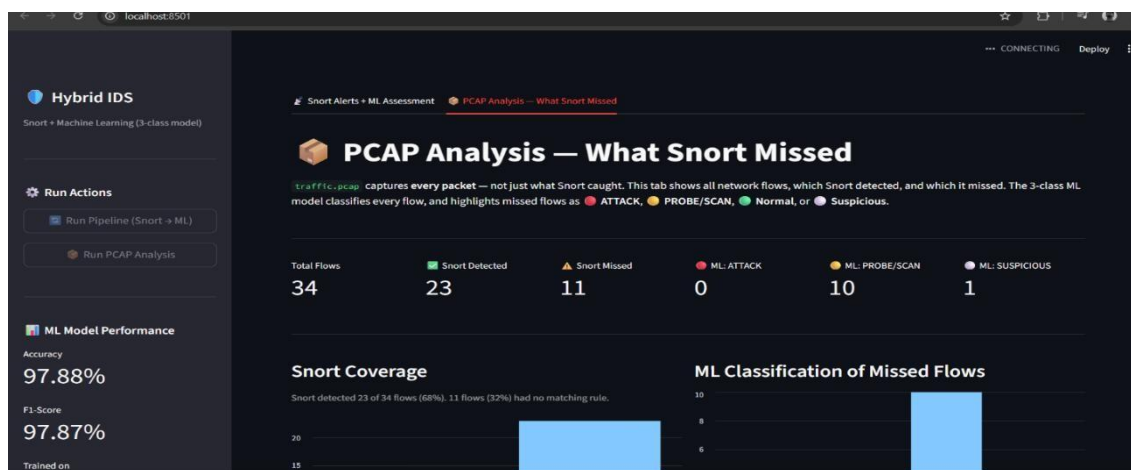


Figure 12: dashboard tab2